

Windows Operating System Terminal Commands — Complete Reference

Command Prompt (cmd.exe) · Built-in Console Tools · PowerShell Essentials

Reference for Windows 10, Windows 11 and Windows Server

A–Z command reference · Multi-command tools · Task-oriented quick index

September 2026

Primary source: Microsoft Learn — Windows commands A–Z reference, PowerShell documentation, and in-box tool documentation for Windows 10/11.

Table of Contents

1. Introduction	1
1.1 Scope and How to Use This Reference	1
1.2 The Windows Command-Line Shells and How to Launch Them	1
1.3 Command Syntax Notation	2
1.4 Getting Help at the Command Line	2
2. Command-Line Fundamentals	2
2.1 Redirection and Pipeline Operators	2
2.2 Wildcards, Environment Variables and ERRORLEVEL	3
2.3 Batch Files in Brief	3
3. A–Z Command Reference	3
A	3
B	4
C	5
D	6
E	8
F	8
G	9
H	10
I	10
J	10
K	10
L	11
M	11
N	12
O	13
P	14
Q	15
R	15
S	17
T	18
U	20
V	20

W	20
X	21
4. Multi-Command Tools and Their Subcommands	21
4.1 diskpart — disk, partition and volume management	21
4.2 net and netdom — users, shares, services and domains	22
4.3 netsh — network configuration shell	23
4.4 sc — Service Control Manager	24
4.5 schtasks — scheduled tasks	24
4.6 reg — registry console operations	25
4.7 fsutil — advanced file system operations	25
4.8 bitsadmin — Background Intelligent Transfer Service	26
4.9 vssadmin — Volume Shadow Copy Service	26
4.10 wbadmin — Windows Server Backup	27
4.11 manage-bde — BitLocker Drive Encryption	27
4.12 diskshadow — VSS scripting interpreter	28
4.13 secedit — local security policy	28
4.14 auditpol — audit policy	28
4.15 certutil — selected operations	29
4.16 nslookup — interactive mode	29
4.17 ftp — interactive commands	30
4.18 telnet — interactive mode	30
5. PowerShell Essentials	31
5.1 Cmdlet Structure and Discovery	31
5.2 Core Cmdlets by Category	31
5.3 CMD-to-PowerShell Equivalent Map	34
6. Task-Oriented Quick Index	35
Appendix A. Redirection, Operators and Batch Symbols	36
Appendix B. Run (Win+R) Dialog and System Utility Commands	37
Appendix C. Compatibility Notes and Sources	39
C.1 Compatibility Notes	39
C.2 Sources	39

1. Introduction

1.1 Scope and How to Use This Reference

This document is a complete, alphabetically ordered reference to the terminal commands built into the Windows operating system. It covers the Win32 console commands available in the Command shell (`cmd.exe`) on Windows 10, Windows 11 and Windows Server; the multi-command interpreters such as `diskpart`, `netsh` and `net`; and a practical introduction to PowerShell cmdlets. Every command is shown with its purpose and a short, typical usage example or its most important switches.

Commands marked “legacy” are retained mainly for compatibility and may be absent on 64-bit systems or replaced by newer tools (the modern replacement is named in the description). Commands marked “Windows Server” or “RSAT/Server” exist primarily on server editions or after installing Remote Server Administration Tools. Tools such as `curl`, `tar`, the OpenSSH client (`ssh`, `scp`, `sftp`, `ssh-keygen`), `wsl` and `winget` are integrated in-box on modern Windows 10/11 releases; the relevant minimum release is noted in the entry.

Many administrative commands require an elevated prompt. Right-click “Command Prompt” or “Windows Terminal” and choose “Run as administrator”; an elevated window title begins with “Administrator”. Use a command only on systems and data you are authorized to manage, and test destructive operations (`format`, `clean`, `delete`, `convert`) on non-production data first.

1.2 The Windows Command-Line Shells and How to Launch Them

Windows ships with two Microsoft command-line shells. The Command shell (`cmd.exe`) runs Win32 console commands and `.bat/.cmd` batch scripts. PowerShell runs the same console commands plus PowerShell cmdlets that pass structured objects through a pipeline; Windows PowerShell 5.1 is built in, and cross-platform PowerShell 7+ runs as `pwsh` when installed. Windows Terminal (`wt.exe`) is the modern tabbed host that can run either shell.

- Press Win+R, type `cmd` and press Enter to open the Command Prompt.
- Press Win+X and choose “Terminal”, “Windows PowerShell” or “Command Prompt” from the quick-link menu.

- In File Explorer, click the address bar, type `cmd` and press Enter to open a prompt in that folder.
- Hold Shift and right-click a folder to obtain “Open in Terminal” / “Open command window here”.
- Type `powershell` or `pwsh` inside any console to switch shells; type `exit` to leave.

1.3 Command Syntax Notation

Examples follow standard Windows notation: square brackets `[ ]` enclose optional items, a vertical bar `|` separates mutually exclusive choices, italic placeholders stand for values you supply, and switches begin with a forward slash (`/switch`) or, for many modern tools, a hyphen (`-switch`). Paths that contain spaces must be quoted, for example `"C:\Program Files"`. Command names are not case-sensitive in `cmd.exe`; PowerShell is also case-insensitive by default for commands and parameters.

1.4 Getting Help at the Command Line

- `help` — list the built-in commands; `help xcopy` shows help for one command.
- `command /?` — display the built-in help, switch list and examples for nearly every command.
- `command /? | more` — page through long help; `command /? > help.txt` saves it to a file.
- In PowerShell: `Get-Help command -Full` (or `help command`), and `Get-Command -Noun Service` to discover commands.

2. Command-Line Fundamentals

2.1 Redirection and Pipeline Operators

The Command shell combines commands with operators. The pipe `|` sends one command’s output to another (`dir | findstr ".pdf"`), angle brackets redirect output to files (`>` overwrites, `>>` appends), and `2>` redirects errors; `2>&1` merges errors into standard output. `&&` chains commands only on success, `||` only on failure, and `&` runs both unconditionally. Appendix A lists every operator and symbol.

2.2 Wildcards, Environment Variables and ERRORLEVEL

Use * to match any sequence of characters and ? for a single character in file names. Environment variables expand between percent signs, such as %PATH%, %USERPROFILE%, %COMPUTERNAME% and %APPDATA%; set displays them, set NAME=value changes one for the session, and setx makes a persistent change. After a command runs, %ERRORLEVEL% holds its numeric exit code (0 usually means success) and can be tested with if errorlevel.

2.3 Batch Files in Brief

A batch file is a plain-text .cmd or .bat script containing commands that run in order. Scripting constructs covered in Section 3 include @echo off, rem (comments), setlocal/endlocal, set /p and set /a, if/else, for loops, call, goto and labels, choice, timeout, pause, shift and exit /b ERRORLEVEL. Save the file as ANSI or UTF-8 text and run it by typing its name; right-click to run as administrator when elevation is required.

3. A–Z Command Reference

Top-level executables are listed alphabetically below; alternative spellings appear on the same row (for example, md (mkdir)). Interactive interpreters and their subcommands (diskpart, net, netsh, sc, schtasks, reg, fsutil, bitsadmin, vssadmin, wbadmin, manage-bde, diskshadow, secedit, auditpol, certutil, nslookup, ftp and telnet) are documented in full in Section 4.

A

Command	Purpose	Example / key switches
adprep	(Windows Server) Prepare an Active Directory forest and domain before adding a domain controller that runs a newer Windows Server version.	adprep /forestprep
append	Legacy: let programs open data files in specified folders as if they were in the current folder. Not supported on 64-bit Windows.	append c:\reports
arp	Display and modify the Address Resolution Protocol (ARP) cache that maps IP addresses to physical MAC addresses.	arp -a

Command	Purpose	Example / key switches
assoc	Display or change the association between a file name extension and a file type.	assoc .txt=txtfile
at	Deprecated: schedule a command at a given time. Use schtasks instead.	at 02:00 /every:M backup.bat
atmadm	Legacy: monitor connections and addresses registered by the ATM (Asynchronous Transfer Mode) network manager.	atmadm
attrib	Display or change file attributes: R read-only, H hidden, S system, A archive, I not content indexed, O offline, U unpinned, P pinned.	attrib +h -r notes.txt
auditpol	Display and configure the system audit policy (categories, subcategories, per-user policy, backups).	auditpol /get /category:*

B

Command	Purpose	Example / key switches
bcdboot	Create or repair the boot environment and system partition boot files (BCD store).	bcdboot C:\Windows /s S: /f UEFI
bcdedit	Display and edit the Boot Configuration Data store that controls Windows startup.	bcdedit /enum bcdedit /set safeboot minimal
bdehdcfg	Prepare a hard drive with the partitions BitLocker Drive Encryption requires.	bdehdcfg -target C: -size 500 -quiet
bitsadmin	Create, download, upload, monitor and administer Background Intelligent Transfer Service (BITS) jobs. Subcommands are listed in Section 4.8.	bitsadmin /transfer job1 /download /priority foreground https://x/f.zip c:\f.zip
bootcfg	Legacy (Windows XP/2003): query and edit boot.ini. Use bcdedit on modern Windows.	bootcfg /list
break	Sets or clears extended CTRL+C checking in CMD; retained for batch compatibility and largely a no-op on Windows.	break on

C

Command	Purpose	Example / key switches
cacls	Deprecated: display or edit access control lists (ACLs). Use icacls.	cacls file.txt /E /G Alice:F
call	Invoke one batch script from another and return to the caller; also used to call a label inside a script.	call tools.bat arg1
cd (chdir)	Display the current directory or change to another one. /d also changes drive.	cd /d D:\projects
certreq	Request, retrieve and accept certificates from a certification authority; build requests from .inf files.	certreq -submit -config CA01\Corp request.csr
certutil	Encode/decode files, dump and verify certificates, manage CA services and certificate stores. Key verbs in Section 4.15.	certutil -hashfile install.iso SHA256
change	Manage Remote Desktop Services host settings: logon (enable/disable sessions), port (COM mapping), user (install/execute mode).	change user /install
chcp	Display or set the active console code page (65001 = UTF-8, 437 = US English, 936 = GBK).	chcp 65001
chdir	Alternative spelling of cd.	chdir ..
chglogon / chgport / chgusr	Short forms of change logon, change port and change user.	chgusr /execute
chkdsk	Check a volume's file system and file system metadata for errors and optionally repair them.	chkdsk C: /f /r /x
chkntfs	Display or modify whether a volume is checked with chkdsk at boot; exclude dirty volumes.	chkntfs /x C: D:
choice	Prompt a user to pick one of a list of choices in a batch script and return the selection as ERRORLEVEL.	choice /C YN /M "Continue"
cipher	Display or alter EFS encryption of files and folders; /w securely wipes free disk space.	cipher /w:C:\temp
cleanmgr	Open or automate the Disk Cleanup tool (preset states via /sageset and /sagerun).	cleanmgr /sagerun:1
clip	Pipe command output or text into the Windows clipboard.	ipconfig /all clip
cls	Clear the console screen.	cls
cmd	Start a new instance of the Windows command interpreter. /c run and exit, /k run and stay, /q quiet, /a /u ANSI/Unicode output.	cmd /k "d: && cd build"

Command	Purpose	Example / key switches
cmdkey	Create, list and delete stored user names, passwords and credentials used for network/remote logon.	cmdkey /list cmdkey /add:server /user:Alice
cmstp	Install or uninstall a Connection Manager service profile (.inf/.cmp).	cmstp /s profile.inf
color	Set console foreground and background colors using two hex digits (0-F).	color 0A
comp	Compare the contents of two files (or sets) byte by byte.	comp /a /l old.txt new.txt
compact	Display or change NTFS file and folder compression.	compact /c /s:C:\data *.docx
convert	Convert a FAT or FAT32 volume to NTFS without reformatting.	convert D: /fs:ntfs /v
copy	Copy one or more files; combine files with +; /y suppresses overwrite prompts, /b binary, /a ASCII.	copy /y a.txt+b.txt merged.txt
cscript	Run scripts through the console-based Windows Script Host (VBScript/JScript).	cscript //nologo script.vbs
curl	Built-in real curl.exe since Windows 10 1803: transfer data over HTTP/HTTPS/FTP and many protocols.	curl -L -o app.zip https://example.com/app.zip

D

Command	Purpose	Example / key switches
date	Display or set the system date; /t prints without prompting.	date /t
dcdiag	(Active Directory) Run diagnostics on a domain controller: DNS, replication, services and advertising.	dcdiag /v /c
dcgpofix	(Active Directory) rebuild the Default Domain Policy or Default Domain Controllers Policy to installation defaults.	dcgpofix /target:domain /ignoreschema
dcpromo	Deprecated: promote/demote a domain controller (up to Server 2008 R2). Use Install-ADDSDomainController on modern servers.	dcpromo /unattend:answer.txt
defrag	Locate and consolidate fragmented files and folders on local volumes.	defrag C: /O /U /V
del (erase)	Delete one or more files. /s recurse, /q quiet, /f force read-only, /p confirm.	del /s /q C:\temp*.tmp
dfsdiag	(DFS) diagnose distributed file system namespaces and referrals (testdcs, testdfsconfig, testreferral, testsites).	dfsdiag /testreferral /dfspath:\\corp\ns

Command	Purpose	Example / key switches
dfsrmig	Migrate SYSVOL replication from FRS to DFSR and report migration state.	dfsrmig /getglobalstate
diantz	Cabinet compression tool; functionally identical to makecab.	diantz data.txt data.cab
dir	List files and subdirectories. /a attributes, /o order, /s recurse, /b bare, /r alternate data streams.	dir /s /b *.pdf
diskcomp	Legacy: compare the contents of two floppy disks.	diskcomp a: b:
diskcopy	Legacy: copy the complete contents of one floppy disk to another.	diskcopy a: b:
diskpart	Interpreter for managing disks, partitions, volumes and virtual disks. Subcommands in Section 4.1.	diskpart /s layout.txt
diskperf	Legacy: enable or disable disk performance counters on remote Windows.	diskperf -y
diskraid	Interpreter for hardware RAID (VDS) storage subsystem configuration.	diskraid /s raid.txt
diskshadow	Interpreter for Volume Shadow Copy Service operations. Subcommands in Section 4.12.	diskshadow /s snap.txt
dism	Deployment Image Servicing and Management: service a running or offline Windows image, enable features, repair component store.	DISM /Online /Cleanup-Image /RestoreHealth
dispdiag	Log display/diagnostics data into a cabinet for graphics troubleshooting.	dispdiag -out display.cab
dnscmd	(DNS Server role) script DNS server, zone and record administration.	dnscmd . /enumzones
doskey	Recall and edit command lines and define reusable command-line macros.	doskey ls=dir /o:gn \$*
driverquery	List installed device drivers with properties, path and startup mode.	driverquery /v /fo csv > drv.csv
dtrace	DTrace for Windows: run tracing scripts against operating system providers (supported builds).	dtrace -n 'proc:::CreateProcess { printf("%s", execname); }'

E

Command	Purpose	Example / key switches
echo	Print a message, expand a variable, or turn command echoing on/off in batch scripts.	@echo off & echo %COMPUTERNAME%
edit	Legacy 16-bit MS-DOS text editor; not present on 64-bit Windows. Use notepad or PowerShell editors.	edit file.txt
endlocal	End the environment localization started by setlocal, restoring prior variables and drive/directory.	endlocal
erase	Alternative spelling of del.	erase old.bak
eventcreate	Write a custom event into an application or system event log.	eventcreate /l application /t error /id 100 /d "Backup failed"
evntcmd	Configure event-to-trap translations and trap destinations from a configuration file.	evntcmd /s sys trapcfg.cnf
exit	Quit CMD or, with /b, exit the current batch script and optionally set ERRORLEVEL.	exit /b 0
expand	Expand one or more compressed update/cabinet files (-F selects members).	expand source.cab -F:* C:\out
extrac32	Extract files from cabinet (.cab) archives; also used by Internet Explorer setup.	extrac32 /y /l C:\out package.cab

F

Command	Purpose	Example / key switches
fc	Compare two files (or sets) and display their differences; /a abbreviated, /b binary, /u Unicode, /n line numbers.	fc /a /n old.txt new.txt
find	Search for a literal text string in one or more files. /v invert, /n line numbers, /i ignore case.	find /i /n "error" app.log
findstr	Powerful text/regex search with wildcards across files; /s recurse, /r regex, /g patterns file, /x exact line.	findstr /s /i /r "failed error [0-9]+"
finger	Legacy: display information about a user on a remote system running the Finger service.	finger Alice@host.example.com
flattemp	Enable or disable flat temporary folders on a Remote Desktop Session Host.	flattemp /enable
fondue	Install optional Windows features (such as .NET Framework 3.5) by downloading from Windows Update.	fondue /enable- feature:NetFx3 /hide- error-output

Command	Purpose	Example / key switches
for	Batch loop over files, numbers, directory trees (/r), parsed text (/f) or token lists (/l).	for %f in (*.log) do echo %f
forfiles	Select files by modification date, name or extension and run a command against each; ideal for retention cleanup.	forfiles /p C:\logs /d -14 /c "cmd /c del @path"
format	Format a volume for FAT/FAT32/exFAT/NTFS/ReFS. /q quick, /fs file system, /p zero passes.	format D: /fs:NTFS /q /y
freedisk	Check whether a volume has the specified free space before continuing a script.	freedisk 1024MB C:
fsutil	Advanced NTFS/ReFS administration: dirty bit, quotas, hard links, reparse points, USN journal, 8.3 names. Section 4.7.	fsutil fsinfo drives
ftp	Interactive and scriptable File Transfer Protocol client. Interactive commands in Section 4.17.	ftp -s:getfiles.txt ftp.example.com
ftype	Display or change the open command associated with a file type (paired with assoc).	ftype txtfile="%SystemRoot%\system32\notepad.exe" %1
fveupdate	Legacy BitLocker/Full Volume Encryption update tool, normally invoked by setup.	fveupdate

G

Command	Purpose	Example / key switches
getmac	Display the MAC address of every network adapter and the transport bound to it.	getmac /v /fo list
gettype	Legacy: detect the Windows product type/role (for example domain controller, server).	gettype /srv
goto	Jump batch execution to a labeled line (:label); supports :eof to end the current call.	goto :eof
gpfixup	Repair Group Policy references and dependency links after a domain rename.	gpfixup /olddns:old.local /newdns:new.local
gpresult	Display Resultant Set of Policy (RSOP) for a user or computer.	gpresult /r /user:Alice gpresult /h rso p.html
gpupdate	Refresh local and Active Directory Group Policy settings.	gpupdate /force /target:computer
graftabl	Legacy: enable an extended character set for graphics-mode (full-screen) programs.	graftabl 437

H

Command	Purpose	Example / key switches
help	List available commands or display help text for a specified command (equivalent to command /?).	help robocopy
helpctr	Legacy Windows XP Help and Support Center launcher.	helpctr
hostname	Print the computer (host) name.	hostname

I

Command	Purpose	Example / key switches
icacls	Display, back up, restore and modify NTFS access control lists; the modern replacement for cacs.	icacls C:\data /grant Alice:(OI)(CI)RX /t
if	Conditional execution in batch scripts: exist, errorlevel, string compare, defined; supports else blocks.	if exist out.txt (echo ready) else (echo missing)
ipconfig	Display and manage IP configuration: /all details, /release, /renew, /flushdns, /displaydns, /registerdns.	ipconfig /all
ipxroute	Legacy: modify and display IPX/SPX routing tables.	ipxroute table
irftp	Legacy: send files over an infrared link through the IrDA stack.	irftp /s photo.jpg

J

Command	Purpose	Example / key switches
jetpack	Compact a WINS or DHCP Jet database for optimal size (Windows Server).	jetpack %SystemRoot%\System32\dhcp\tmp.mdb dhcp.mdb

K

Command	Purpose	Example / key switches
klist	Display cached Kerberos tickets and purge them; query session tickets and Key Distribution Center bindings.	klist purge
ksetup	Configure Kerberos realms, KDC hosts and realm flags for a computer not joined to a domain.	ksetup /setrealm EXAMPLE.COM

Command	Purpose	Example / key switches
ktmutil	Kernel Transaction Manager utility; recover or list Transacted File System (TxF) transactions.	ktmutil txf recover C:
ktpass	Configure a non-Windows service's principal name and export a Kerberos keytab (.keytab).	ktpass -out svc.keytab -princ svc/host@REALM -mapuser DOM\svc -crypto AES256-SHA1 -ptype KRB5_NT_PRINCIPAL

L

Command	Purpose	Example / key switches
label	Create, change or delete the volume label of a disk.	label D: Archive
lodctr	Register or rebuild performance counter name strings from an .ini file; /r rebuilds from backup.	lodctr /r
logman	Create and manage Event Tracing sessions and performance counter logs/alerts.	logman create trace trace1 -p "Windows Kernel Trace" -o trace.etl
logoff	Log off a user or session, locally or on a Remote Desktop Session Host.	logoff 2 /server:RDS01
lpq	Query the status of an LPD print queue.	lpq -S printsrv -P laser
lpr	Send a print job to a printer or print server running LPD/LPR.	lpr -S printsrv -P laser report.pdf

M

Command	Purpose	Example / key switches
macfile	Legacy (Windows Server): manage File Server for Macintosh volumes, directories and files.	macfile volume /add
makecab	Compress files into a cabinet (.cab); alias diantz.	makecab /D CompressionType=LZX data.txt data.cab
manage-bde	Turn BitLocker on/off, configure protectors, lock/unlock and report status. Section 4.11.	manage-bde -status C:
mapadmin	Manage User Name Mapping for Services for NFS (Windows Server).	mapadmin list
md (mkdir)	Create a directory or a directory tree.	md C:\proj\src\core
mkdir	Alternative spelling of md.	mkdir backup
mklink	Create a symbolic link (/D directory), hard link (/H) or directory junction (/J).	mklink /D C:\link C:\real\folder

Command	Purpose	Example / key switches
mmc	Launch a blank Microsoft Management Console or open an .msc console; /a author mode.	mmc services.msc
mode	Configure serial (COM) ports, console window size, keyboard rate, code-page status and device status.	mode con cols=140 lines=45
more	Display output one screen at a time; +n start line, /e extended, /s squeeze blank lines.	type huge.log more +100
mountvol	Create, list or delete volume mount points and GUID volume paths; no drive letter required.	mountvol C:\mnt \\?\Volume{GUID}\
move	Move one or more files and rename directories; /y suppresses prompts.	move /y *.log C:\archive
mqbkup	Back up and restore MSMQ message storage files and registry keys.	mqbkup -b C:\mqbak
mqsvc / mqtgsvc	Run the Message Queuing service (mqsvc) or the MSMQ Triggers service (mqtgsvc) from the command line.	mqsvc
msdt	Launch a Microsoft Support Diagnostic Pack to troubleshoot a known area.	msdt -id WindowsUpdateDiagnostic
msg	Send a pop-up message to a user, session or all sessions on an RDS host.	msg * /server:RDS01 "Reboot in 5 minutes"
msiexec	Install, repair, patch, advertise or uninstall Windows Installer .msi/.msp packages.	msiexec /i setup.msi /qn /l*v install.log
msinfo32	Open System Information; export a full inventory with /report or /nfo.	msinfo32 /report C:\sysinfo.txt
mstsc	Start Remote Desktop Connection; /v target, /admin console session, /edit or /w /h geometry.	mstsc /v:RDS01 /admin

N

Command	Purpose	Example / key switches
nbtstat	NetBIOS-over-TCP/IP statistics, name cache, sessions and name registration; -A resolve by IP, -RR refresh.	nbtstat -A 10.0.0.5
net	Umbrella tool for users, groups, shares, sessions, services, accounts and mapped drives. Subcommands in Section 4.2.	net use Z: \\server\share /persistent:yes
netcfg	Install, preview, uninstall or query Win32 networking components.	netcfg -v -u ms_tcpip

Command	Purpose	Example / key switches
netdom	(RSAT/Server) join/remove domains, rename computers, reset machine passwords and manage trusts (Section 4.2).	netdom join PC01 /domain:corp /userd:admin /passwordd:*
netsh	Network configuration interpreter for interfaces, firewall, WLAN, WinHTTP proxy, winsock and more. Section 4.3.	netsh wlan show profiles
netstat	Display active connections, listening ports, Ethernet stats and the routing table; -b owning process, -n numeric, -o PID.	netstat -ano findstr :443
nfsadmin	Manage Server for NFS or Client for NFS (Services for NFS).	nfsadmin server
nfsshare	Create, list, remove and configure NFS shares.	nfsshare share=D:\share
nfsstat	Display or reset NFS protocol and call statistics.	nfsstat /r
nlbmgr	(Windows Server) open the Network Load Balancing Manager.	nlbmgr
nltest	Test Active Directory secure channels, domain-controller locator results, trusts and replication flags.	nltest /dsgetdc:corp.local
nslookup	Query DNS interactively or non-interactively. Interactive commands in Section 4.16.	nslookup -type=mx example.com 8.8.8.8
ntbackup	Legacy backup utility (Windows XP/2003). Modern equivalent: wbadmin / Windows Server Backup.	ntbackup backup C: /f backup.bkf
ntcmdprompt	Legacy: start CMD inside a 16-bit MS-DOS subsystem (command.com) session.	ntcmdprompt
ntfrsutl	Inspect and manage the legacy NT File Replication Service (FRS) and its ID table/journal.	ntfrsutl sets

O

Command	Purpose	Example / key switches
openfiles	List or disconnect files and folders opened by network users; /local enables the local maintain-objects-list flag.	openfiles /query /v openfiles /disconnect /id 12

P

Command	Purpose	Example / key switches
pagefileconfig	Legacy scripted page-file configuration; modern equivalent is wmic/PowerShell (Set-CimInstance Win32_PageFileUsage).	pagefileconfig /query
path	Display or set the PATH search path for executable files for the current session.	path %PATH%;C:\tools
pathping	Combine ping and tracert: trace a route and report packet loss at every router over time.	pathping -n host.example.com
pause	Suspend a batch script and display "Press any key to continue . . ."	pause
pbadmin	Legacy Phone Book Administrator for Connection Point Services.	pbadmin phonebook.pbk
pentnt	Ancient utility that detected the Pentium floating-point division flaw.	pentnt
perfmon	Open Performance Monitor; /report generates a System Diagnostics report.	perfmon /res
ping	Send ICMP echo requests to test reachability, name resolution and latency.	ping -n 4 -l 1024 host
pktmon	In-box packet monitor and network sniffer (Windows 10 1809+): capture, filter and convert to text/pcapng.	pktmon start --capture --comp all --p 0 -f trace.etl
pnpunattend	Audit a computer for drivers and perform unattended online driver installation.	pnpunattend /auditSystem /install
pnputil	Enumerate devices and third-party driver packages; add/delete drivers, scan for hardware changes.	pnputil /enum-drivers
popd	Return to the directory last saved with pushd (also removes its temporary drive mapping).	popd
powershell	Launch Windows PowerShell 5.1; -Command run code, -File run a script, -NoProfile skip profile.	powershell -NoProfile -Command "Get-Service Where Status -eq Running"
powershell_ise	Launch the PowerShell Integrated Scripting Environment (not shipped with PowerShell 7).	powershell_ise script.ps1
print	Send a text file to a local or network printer.	print /d:\\server\printer page.txt
prncnfg	Configure an existing printer (name, port, sharing, paper).	prncnfg -t -p "Laser" -h "Upstairs printer"
prndrvr	Add, delete or list printer drivers.	prndrvr -a -m "Generic / Text Only"
prnjobs	Pause, resume, cancel or list print jobs.	prnjobs -l

Command	Purpose	Example / key switches
prnmngr	Add/delete/list printers and default printer connections.	prnmngr -a -p "Laser" -m "Driver" -r "IP_10.0.0.9"
prnport	Create/delete/list TCP/IP printer ports.	prnport -a -r IP_10.0.0.9 -h 10.0.0.9
prnqctl	Pause/resume/purge a printer or print a test page.	prnqctl -e -p "Laser"
prompt	Customize the command prompt text (\$p path, \$g >, \$t time, \$\$ dollar, \$c/\$f brackets).	prompt [\$p]\$_\$\$
pubprn	Publish a printer to Active Directory.	pubprn "Laser" LDAP://CN=Computers,DC=corp,DC=local
pushd	Save the current directory and switch to the target; a UNC path is auto-mapped to a free drive letter.	pushd \\server\share
pushprinterconnections	Deploy deployed-printer Group Policy connections at user logon.	pushprinterconnections
pwllauncher	Enable or disable Windows To Go startup options (boot from USB).	pwllauncher /enable
pwsh	Launch cross-platform PowerShell 7+ when installed (separate from Windows PowerShell 5.1).	pwsh -c "\$PSVersionTable.PSVersion"

Q

Command	Purpose	Example / key switches
qappsrv	Legacy: list Remote Desktop Session Host servers on the network.	qappsrv
qprocess	Display information about processes and sessions (alias of query process).	qprocess *
query	Query Remote Desktop Services: process, session, termserver and user information.	query session
quser	Display users logged on to an RDS host (alias of query user).	quser /server:RDS01
qwinsta	Display session/station information (alias of query session).	qwinsta

R

Command	Purpose	Example / key switches
rd (rmdir)	Remove a directory; /s removes the whole tree, /q without confirmation.	rd /s /q C:\build
rdpsign	Digitally sign an .rdp file so clients trust its settings.	rdpsign /sha1 <thumbprint> connect.rdp

Command	Purpose	Example / key switches
recover	Salvage readable data from a disk that contains defective sectors.	recover damaged.txt
refsutil	Repair and diagnose ReFS volumes: salvage, compression, dedup, fixboot, leak and iometrics.	refsutil salvage -D -x C:\recovered C:
reg	Query, add, delete, copy, import/export, load and back up registry keys. Subcommands in Section 4.6.	reg query HKLM\SOFTWARE /s /f "Windows"
regini	Apply registry values, keys and ACL changes described in a text script.	regini permissions.txt
regsvr32	Register or unregister a COM/ActiveX control (.dll/.ocx); /s silent, /u unregister.	regsvr32 /u oldlib.dll
relog	Resample or convert performance logs between blg, csv, tsv and sql formats.	relog perf.blg -f csv -o perf.csv
rem	Mark a comment line in a batch script (:: is an alternative inside labels).	rem --- backup section ---
ren (rename)	Rename one or more files (supports wildcards).	ren *.jpeg *.jpg
repadmin	(Active Directory) diagnose and force replication, show metadata, queue and outbound partners.	repadmin /showrepl * /errorsonly
replace	Replace or add files in a target folder; /a add missing only, /s subfolders, /u newer only.	replace /u new.dll C:\app
reset session	Reset (terminate) an RDS session by ID.	reset session 3 /server:RDS01
rexec	Legacy/insecure: run a command on a host running the rexec service. Use SSH or winrs.	rexec host -l user command
risetup	Legacy Remote Installation Services setup command.	risetup
rmdir	Alternative spelling of rd.	rmdir emptyfolder
robocopy	Robust, restartable file and directory copy with mirroring, retries, logging and thousands of switches.	robocopy C:\src D:\dst /MIR /Z /R:2 /W:2 /LOG:copy.log
route	Display or edit the local IP routing table: print, add, delete, change.	route -p add 10.10.0.0 mask 255.255.0.0 192.168.1.1
rpcinfo	List RPC programs registered on a remote host (part of Services for NFS tools).	rpcinfo -p nfsserver
rpcping	Test RPC connectivity and binding to an RPC server.	rpcping -t ncacn_ip_tcp -s server
rsh	Legacy/insecure: run commands on a remote rsh host. Prefer SSH or winrs.	rsh host -l user command
rundll32	Invoke an exported function of a system DLL (control-panel applets, lock workstation, copy-profile).	rundll32 user32.dll,LockWorkStation

Command	Purpose	Example / key switches
rwinsta	Reset a session's hardware/software state (alias of reset session).	rwinsta 2

S

Command	Purpose	Example / key switches
san	Set the storage area network (SAN) disk online/offline policy for newly detected disks.	san policy=OfflineAll
sc	Service Control: query, configure, create, delete, start/stop and set failure actions for services. Section 4.4.	sc query type= service state= all
schtasks	Create, query, change, run, end and delete scheduled tasks. Section 4.5.	schtasks /create /tn backup /tr C:\b.bat /sc daily /st 02:00
scwcmd	(Windows Server) command-line front end for the Security Configuration Wizard (analyze/configure/transform).	scwcmd analyze /p:policy.xml
scp	Built-in OpenSSH client (Windows 10 1809+): securely copy files between hosts over SSH.	scp file.txt Alice@host:/home/alice/
secedit	Analyze, configure, export, import and validate local security policy. Section 4.13.	secedit /export /cfg policy.inf
serverceipoptin / serverweroptin	Opt in or out of the Customer Experience Improvement Program / Windows Error Reporting.	serverweroptin /disable
set	Display, set or remove environment variables; /a arithmetic, /p prompt for input.	set /a total=2*(3+4)
setlocal	Begin localization of environment changes (optionally ENABLEDELAYEDEXPANSION); ended by endlocal.	setlocal enabledelayedexpansion
setspn	Read, add, remove or reset service principal names (SPNs) for Active Directory accounts.	setspn -L web01\$
setx	Persistently set a user (/M machine-wide) environment variable in the registry (future sessions).	setx EDITOR "C:\tools\nvim.exe" /M
sfc	System File Checker: scan and restore protected Windows system files.	sfc /scannow
sftp	Built-in OpenSSH secure FTP interactive client (Windows 10 1809+).	sftp Alice@host
shadow	Remotely view or control another user's Remote Desktop session.	shadow 1 /control

Command	Purpose	Example / key switches
shift	Shift positional batch parameters (%1, %2 ...) left; supports /n start position.	shift /2
showmount	List directories exported by an NFS server.	showmount -e nfsserver
shutdown	Shut down, restart, log off, hibernate or abort a pending shutdown, locally or remotely.	shutdown /r /t 60 /c "Reboot" shutdown /a
sort	Sort input text; /r reverse, /+n column, /o output file.	sort names.txt /o sorted.txt
ssh	Built-in OpenSSH client (Windows 10 1809+): secure remote shell and tunneling.	ssh -p 22 Alice@host
ssh-keygen	Generate, convert and rotate SSH authentication key pairs (ed25519, rsa, ecdsa).	ssh-keygen -t ed25519 -C "laptop"
start	Launch a program, document or URL in a new window; /b same window, /min /max, /wait, /affinity.	start "" /max notepad readme.txt
subst	Associate a path with a virtual drive letter, or remove the association (/d).	subst Z: C:\work subst Z: /d
sxstrace	Trace and parse WinSxS side-by-side assembly activation failures.	sxstrace trace /logfile:sxs.etl
sysmon	System Monitor service/tool (Sysinternals, when installed) that logs process, network and registry activity.	sysmon -accepteula -i sysmonconfig.xml
sysocmgr	Legacy Windows Optional Component Manager (XP/2003).	sysocmgr /i:sysoc.inf
systeminfo	Display OS version, install date, boot time, hotfixes, processor and memory for the local or a remote host.	systeminfo /s RDS01 /fo list

T

Command	Purpose	Example / key switches
takeown	Take ownership of a file or folder so an administrator can change its ACL; /r recurse.	takeown /f C:\protected /r /d y
tapicfg	Create, remove or publish TAPI application directory partitions.	tapicfg publish
taskkill	Terminate processes by image name or PID; /f force, /t tree, /s remote system, /im image name.	taskkill /f /im chrome.exe /t
tasklist	List running processes with PID, session and memory; /svc services per process, /m loaded modules, /v verbose.	tasklist /svc /fi "imagename eq svchost.exe"
tcmsetup	Set up or disable the TAPI client for a telephony server.	tcmsetup /c tapi01

Command	Purpose	Example / key switches
telnet	Telnet client with an interactive command mode (Section 4.18); installed as an optional feature.	telnet host 23
tftp	Legacy simple file transfer client to a TFTP server.	tftp -i host get firmware.bin
time	Display or set the system time; /t prints without prompting.	time /t
timeout	Pause a script for a number of seconds, allowing a key press to skip; /nobreak ignores keys.	timeout /t 30 /nobreak
title	Set the title of the current console window.	title Build runner
tlntadm	Administer the legacy Telnet Server (sessions, port, authentication).	tlntadm -p 23
tpmtool	Retrieve TPM device information, readiness and owner-auth details.	tpmtool getdeviceinformation
tpmvscmgr	Create and temporarily manage TPM-backed virtual smart cards.	tpmvscmgr create /name "Dev VSC" /adminkey DEFAULT
tracert	Convert Event Tracing (.etl/.blg) logs into CSV/XML summary reports.	tracert trace.etl -o report.xml -of XML
tracert	Trace the IP route to a destination, showing each hop and latency.	tracert -d -h 20 host
tree	Draw the folder tree graphically; /f include files, /a ASCII characters.	tree /f /a more
tscon	Connect a user session to an RDS session (optionally with a password).	tscon 1 /dest:console
tsdiscon	Disconnect a Remote Desktop Services session.	tsdiscon 2
tsecimp	Import TAPI security assignment information from an XML file.	tsecimp /f tapi.xml
tskill	End a process inside an RDS environment by process or session ID.	tskill notepad /server:RDS01
tsprof	Copy or update Remote Desktop Services user profile configuration.	tsprof /copy domain\old domain\new
type	Display the contents of one or more text files.	type a.txt b.txt
typeperf	Stream performance counter values to the screen or a log at a chosen interval.	typeperf "\Processor(_Total)\%% Processor Time" -sc 10
tzutil	List (/l), get (/g) or set (/s) the system time zone; /d disable DST adjustments.	tzutil /s "China Standard Time"
tar	Built-in bsdtar since Windows 10 17063: create/extract .tar, .tar.gz and .zip archives.	tar -xf archive.zip -C C:\out

U

Command	Purpose	Example / key switches
unlodctr	Remove performance counter text and explain text for a driver or service.	unlodctr MyService

V

Command	Purpose	Example / key switches
ver	Display the operating system version number.	ver
verifier	Open Driver Verifier Manager to stress-test and fault-inject drivers.	verifier /query
verify	Toggle whether CMD verifies that files are written correctly to disk (on/off).	verify on
vol	Display the volume label and serial number of a disk.	vol C:
vssadmin	Administer Volume Shadow Copy Service: writers, providers, shadows and storage. Section 4.9.	vssadmin list writers

W

Command	Purpose	Example / key switches
waitfor	Pause until a named system signal is received locally or sent across machines; useful to sync scripts.	waitfor /s host BuildDone
wbadmin	Windows Server Backup command line: enable backup, back up volumes/system state, recover and catalog. Section 4.10.	wbadmin start backup - backuptarget:E: - include:C: -quiet
wdsutil	Administer Windows Deployment Services: add images, configure the server and transport.	wdsutil /get-server /show:all
wecutil	Configure the Windows Event Collector service and event subscriptions.	wecutil es
wevtutil	Query, export, clear and register event logs and publishers.	wevtutil qe System /c:20 /rd:true /f:text
where	Locate files matching a pattern in PATH (or recursively with /r); lists every match in order.	where /r C:\ python.exe
whoami	Print the current user; /upn, /fqdn, /logonid, /groups, /priv, /all show identity details.	whoami /priv
winnt / winnt32	Legacy 16/32-bit Windows setup programs used for in-place installation.	winnt32 /unattend:unattend.txt

Command	Purpose	Example / key switches
winrs	Run a command remotely through WinRM (Windows Remote Management).	winrs - r:https://server:5986 -u admin cmd
winsat	Windows System Assessment Tool: formal assessment or targeted tests (mem, disk, cpu, d3d, mfmedia).	winsat formal
wmic	Deprecated WMI command line; use PowerShell CIM cmdlets. Still present for compatibility.	wmic os get caption,buildnumber
wscript	Run VBScript/JScript through the GUI-based Windows Script Host.	wscript //B script.vbs
wsl	Manage the Windows Subsystem for Linux: install, list distributions, run commands, set defaults.	wsl --install -d Ubuntu wsl -l -v
winget	Windows Package Manager client: search, install, upgrade and uninstall software from configured sources.	winget upgrade --all -- accept-source-agreements

X

Command	Purpose	Example / key switches
xcopy	Extended copy of files and directory trees with attributes and date filters; robocopy is recommended for new scripts.	xcopy C:\src D:\dst /e /h /r /k /y

4. Multi-Command Tools and Their Subcommands

The tools in this section are interpreters or umbrella commands: the first token selects the tool and the following token selects an action. Run the tool with `/?` (or `help` inside it) for complete, version-specific syntax.

4.1 diskpart — disk, partition and volume management

Run `diskpart` in an elevated command prompt. Select an object first (select `disk/partition/volume/vdisk`); later commands act on the selected object. Script it with `diskpart /s file.txt`.

Subcommand	Purpose
list disk / list volume / list partition	Enumerate disks, volumes or partitions with numbers and sizes.
select disk volume partition vdisk n	Focus subsequent commands on the chosen object.
detail disk volume partition vdisk	Show properties of the selected object.

Subcommand	Purpose
clean [all]	Remove partition formatting (all writes zeroes across the disk).
convert gpt mbr basic dynamic	Convert an empty disk's partition style or disk type.
create partition primary efi extended logical msr	Create a partition of the given type and size.
create volume simple mirror raid stripe	Create a volume across one or more disks.
assign letter=L assign mount=	Give the selected volume a drive letter or mount folder.
active / inactive	Mark the selected partition as (not) the active bootable partition (MBR).
format fs=ntfs label=Data quick	Format the selected volume.
extend [size=n] disk=n	Grow the selected volume into adjacent unallocated space.
shrink desired=n querymax	Shrink the selected volume to free space.
delete disk partition volume [override]	Delete the selected object.
online offline disk volume	Bring an object online or take it offline.
attributes disk volume	Show/set read-only and hidden attributes (set readonly).
create vdisk / attach vdisk / detach vdisk	Create, mount or unmount a .vhd/.vhdx virtual disk.
expand vdisk / merge vdisk / compact vdisk	Resize, merge a differencing disk or compact a VHDX.
rescan	Rescan I/O buses for newly attached disks.
automount enable disable scrub	Control automatic mounting of new basic volumes.
uniqueid disk [id=]	Show or change the MBR signature / GPT disk identifier.
gpt [attributes=]	Show or assign GPT attributes on the selected partition.
filesystems	Show file systems supported for formatting the selected volume.
retain	Prepare a simple volume for use as a boot volume on MBR/GPT.
remove letter=L [dismount]	Remove a drive letter or mount point.
import vdisk	Import a virtual disk for use.
exit	Leave the diskpart interpreter.

4.2 net and netdom — users, shares, services and domains

The net family manages local and network resources. net help <command> shows full syntax. netdom (RSAT/Server) handles Active Directory machine and trust operations.

Subcommand	Purpose
net accounts	Set password and logon policy (minimum length, age, lockout).
net computer \\host /add /del	Add or remove a computer account in the domain.
net config workstation server	Show or change workstation/server service settings.
net continue / net pause service	Resume or pause a Windows service.
net file [id] [/close]	List or close shared files opened by remote users.

Subcommand	Purpose
net group / net localgroup	Manage domain global groups or local groups (add, delete, members).
net helpmsg nnnn	Explain a Windows network/system error number.
net send / net name	Legacy Messenger service messaging and message aliases.
net session	List or disconnect sessions with the local server.
net share	List, create or delete network shares (net share Data=D:\data /grant:Alice,full).
net start / net stop service	Start or stop a service (also net start lists running services).
net statistics workstation server	Show protocol/service statistics.
net time	Display or synchronize the computer clock (modern: w32tm).
net use	Map/disconnect network drives and printer ports; net use * /del /y clears all.
net user	Create, delete, enable and set passwords for local/domain user accounts.
net view	List shared resources on a computer or domain.
netdom join / remove	Join a domain/workgroup or remove the computer from it.
netdom computername	Add/remove/make-primary computer names.
netdom resetpwd	Reset a domain controller's machine account password.
netdom trust	Create, verify or reset trusts between domains.
netdom query	Query workstation, server, DC, OU, pdc or fsmo information.
netdom verify / reset	Verify or reset the secure channel to a domain.

4.3 netsh — network configuration shell

netsh is context-based: netsh <context> <command>. Run netsh to enter interactive mode; contexts include interface, advfirewall, wlan, winhttp, winsock, dnsclient, dhcpclient, http, lan, mbn, ras, trace and wfp.

Subcommand	Purpose
netsh interface show interface	List adapter names and state.
netsh interface ip set address name static IP mask gw	Assign a static IPv4 address, mask and gateway.
netsh interface ip set dns name static addr	Set the primary DNS server (index=2 adds more).
netsh interface ip reset log.txt	Reset TCP/IP stack settings to defaults.
netsh wlan show profiles	List saved Wi-Fi profiles.
netsh wlan show profile name=X key=clear	Reveal a saved Wi-Fi password (elevated).
netsh advfirewall firewall add rule	Create an inbound/outbound firewall rule (name, dir, action, protocol, localport).
netsh advfirewall set allprofiles state off on	Turn the firewall off or on per/all profiles.
netsh winhttp set proxy proxy:8080 bypass-list	Configure the system (WinHTTP) proxy used by services.
netsh winhttp import proxy source=ie	Copy proxy settings from Internet Explorer/Edge.

Subcommand	Purpose
netsh winsock reset	Reset the Winsock catalog (reboot required).
netsh dnsclient show state set search	Display global DNS settings or configure the DNS search list; clear the resolver cache with ipconfig /flushdns.
netsh trace start capture=yes	Start a packet/network trace; stop with netsh trace stop.
netsh dump / exec file.txt	Export the running configuration / re-apply a saved script.
netsh -c context -f file / -r host	Run a script file or execute against a remote host.

4.4 sc — Service Control Manager

sc communicates with Service Control Manager. Note the required space after the equals sign in flags such as start= auto.

Subcommand	Purpose
sc query [name] / queryex	List service state (queryex adds PID, flags and exit codes).
sc qc name	Show a service's configuration (binPath, start type, account).
sc start / stop / pause / continue name	Control service lifecycle.
sc config name start= demand auto disabled	Change startup type and other settings (binPath=, obj=, depend=).
sc create name binPath= start=	Register a new service.
sc delete name	Remove a service registration.
sc description name text	Set the service description.
sc failure name reset= actions=	Define restart actions when a service fails.
sc sdshow / sdset name	Show or set a service's security descriptor (SDDL).
sc getdisplayname / getkeyname	Translate between service key name and display name.
sc enumdepend name	List services that depend on the named service.
sc boot ok bad	Mark the last boot as good or bad.
sc lock / querylock	Lock the SCM database or query lock state.

4.5 schtasks — scheduled tasks

schtasks replaces the legacy at command and manages Task Scheduler jobs from scripts. Tasks are identified by /tn name.

Subcommand	Purpose
schtasks /create	Create a task: /sc schedule (minute hourly daily weekly monthly once onlogon onstart), /st start time, /tr program, /ru account, /rl highest, /f overwrite.
schtasks /query	List tasks; /v verbose, /fo csv table list, /tn one task.
schtasks /change	Modify a task: program (/tr), user, enable/disable, password.
schtasks /run / /end	Start a task now or stop a running instance.

Subcommand	Purpose
<code>schtasks /delete /f</code>	Delete a task without confirmation.
<code>schtasks /showsid</code>	Show the security principal SID a task runs as.

4.6 reg — registry console operations

Registry roots: HKLM, HKCU, HKCR, HKU, HKCC. Back up a key before editing (reg save/export).

Subcommand	Purpose
<code>reg query key [/v value]/s/f pattern]</code>	Read values, recurse subkeys or search for data.
<code>reg add key /v name /t type /d data /f</code>	Create or overwrite a value (REG_SZ, REG_DWORD, REG_EXPAND_SZ...).
<code>reg delete key [/v name] /f</code>	Delete a key or value.
<code>reg copy src dst /s /f</code>	Copy a registry branch.
<code>reg save key file.hiv / reg restore key file.hiv</code>	Save/restore a binary hive backup.
<code>reg export key file.reg / reg import file.reg</code>	Export/import text .reg files.
<code>reg load / unload key file.hiv</code>	Mount or release an offline hive for editing.
<code>reg compare k1 k2</code>	Compare two registry keys and report differences.

4.7 fsutil — advanced file system operations

Most fsutil commands require an elevated prompt. Resource: `fsutil <object> /?` lists object-specific syntax.

Subcommand	Purpose
<code>fsutil fsinfo drives volumeinfo ntfsinfo</code>	Enumerate drives and report file-system details.
<code>fsutil dirty query set C:</code>	Query or set the volume dirty bit (forces chkdsk at boot).
<code>fsutil behavior query set</code>	Tune NTFS behavior (8dot3 names, SymlinkEvaluation, memory usage).
<code>fsutil 8dot3name query strip set</code>	Manage legacy 8.3 short file names.
<code>fsutil file createnew queryallocated setzerodata</code>	Create fixed-size files, inspect ranges or zero data.
<code>fsutil hardlink create</code>	Create an NTFS hard link.
<code>fsutil reparsepoint query delete</code>	Inspect or remove reparse points (symbolic links, junctions).
<code>fsutil quota track disable modify query</code>	Manage per-user disk quotas.
<code>fsutil objectid query set delete</code>	Manage object identifiers used by distributed links.
<code>fsutil repair query set wait C:</code>	Manage NTFS self-healing repair state.
<code>fsutil sparse setflag queryflag setrange</code>	Manage sparse files.
<code>fsutil usn readjournal createjournal queryjournal</code>	Work with the NTFS change (USN) journal.
<code>fsutil volume diskfree allocationreport filelayout</code>	Report free space, cluster usage and file layout.
<code>fsutil resource / transaction</code>	Manage Transactional NTFS (TxF) resource managers and transactions.

Subcommand	Purpose
fsutil devdrv query enable disable	Manage Dev Drive (developer performance volumes).
fsutil wim	Manage WIM-backed (cloud) files.

4.8 bitsadmin — Background Intelligent Transfer Service

Typical lifecycle: create -> addfile -> (set credentials/flags) -> resume -> monitor state TRANSFERRED -> complete. The convenience verb transfer does it all in one line.

Subcommand	Purpose
bitsadmin /create name	Create an upload or download job.
bitsadmin /addfile name url local	Add a remote/local file pair to the job.
bitsadmin /resume /suspend name	Start or pause transfer (new jobs are suspended).
bitsadmin /complete /cancel name	Finalize a transferred job or cancel it.
bitsadmin /list /listfiles /monitor	Enumerate jobs, their files, or watch progress.
bitsadmin /getstate /getprogress name	Read job state and byte progress.
bitsadmin /setpriority name foreground high normal low	Set transfer priority.
bitsadmin /setcredentials name target scheme user pass	Attach authentication credentials.
bitsadmin /transfer name url local	One-shot synchronous download/upload.
bitsadmin /reset /util /version	Cancel all jobs for the owner or report BITS version.

4.9 vssadmin — Volume Shadow Copy Service

Requires an elevated prompt. Used to inspect VSS writers/providers and manage shadow copies and shadow-storage space.

Subcommand	Purpose
vssadmin list writers	List VSS writers and their state (backup troubleshooting).
vssadmin list providers shadows shadowstorage	List providers, existing snapshots or storage allocation.
vssadmin create shadow /for=C:	Create a shadow copy of a volume.
vssadmin delete shadows /for=C: /oldest /all	Delete snapshots by volume, age or ID.
vssadmin resize shadowstorage /on=C: /for=C: /maxsize=10%%	Resize the area reserved for snapshots.
vssadmin add shadowstorage	Allocate shadow storage on a different volume.

4.10 wadmin — Windows Server Backup

wadmin is the command-line interface for Windows Server Backup (and system recovery on Windows clients).

Subcommand	Purpose
wadmin enable backup / disable backup	Configure or remove the scheduled backup.
wadmin start backup -backuptarget -include	Run a one-time backup of volumes or items.
wadmin start systemstatebackup	Back up Active Directory/system state (Server).
wadmin start recovery / start sysrecovery	Recover files, volumes, apps or the whole system.
wadmin start systemstaterecovery	Perform authoritative/non-authoritative system-state restore.
wadmin get versions items disks status	List recovery points, contained items, target disks or job status.
wadmin delete catalog delete systemstatebackup	Remove the backup catalog or old system-state versions.
wadmin restore catalog	Rebuild the catalog from a backup target when it is corrupted.
wadmin stop job	Cancel the running backup or recovery.

4.11 manage-bde — BitLocker Drive Encryption

Controls BitLocker from an elevated prompt. protectors include TPM, recovery password, startup key and password.

Subcommand	Purpose
manage-bde -status C:	Show encryption status, protectors and conversion progress.
manage-bde -on C: -rp	Turn BitLocker on and add a numerical recovery password.
manage-bde -off C:	Decrypt and turn BitLocker off.
manage-bde -pause -resume C:	Pause or resume encryption/decryption.
manage-bde -lock -unlock C:	Lock a data volume or unlock it with -pw/-rk.
manage-bde -autounlock -enable -disable C:	Allow automatic unlocking of a data volume.
manage-bde -protectors -add -get -delete C:	Manage TPM, password, recovery key and USB protectors.
manage-bde -changepassword -changePIN -changekey	Rotate password, PIN or startup key.
manage-bde -forcerecovery C:	Force recovery mode at next restart.
manage-bde -keypackage -backup	Export a recovery key package or back the recovery key to AD.
manage-bde -wiperecovery C:	Wipe free space after encryption for data sanitization.
manage-bde -tpm	Configure TPM ownership and state.

4.12 diskshadow — VSS scripting interpreter

diskshadow is the scriptable shadow-copy engine used by backup applications; it supports interactive and /s script mode.

Subcommand	Purpose
set context persistent volatile	Choose whether snapshots survive the session.
set metadata meta.cab	Define where writer metadata is stored.
add volume C: alias MySnap	Include a volume in the shadow set with an alias.
begin backup / create / end backup	Create the shadow copy inside a backup session.
list writers providers shadows	Enumerate VSS writers, providers and copies.
load metadata meta.cab	Load writer metadata for a restore.
begin restore / ... / end restore	Run a restore session sequence.
expose %alias% X:	Expose a snapshot as a drive letter.
unexpose X:	Remove an exposed snapshot.
reset	Clear the current shadow-set state.
delete shadows	Delete shadow copies by id/volume/oldest/all.
simulate restore / revert	Validate a restore or revert a volume to a shadow copy.

4.13 secedit — local security policy

Automates Security Settings extension operations against .inf policy templates and the local security database.

Subcommand	Purpose
secedit /analyze /db secedit.sdb /cfg policy.inf	Compare current policy against a template.
secedit /configure /db /cfg	Apply a policy template to the computer.
secedit /export /cfg out.inf	Export effective local policy to text.
secedit /import /cfg in.inf /db	Import a template into a security database.
secedit /generaterollback	Create a rollback template before applying policy.
secedit /validate file.inf	Validate a security template's syntax.

4.14 auditpol — audit policy

Controls category and subcategory auditing and per-user audit policy from an elevated prompt.

Subcommand	Purpose
auditpol /get /category:*	List every audit category and subcategory setting.
auditpol /set /subcategory:"Logon" /success:enable	Enable success/failure auditing for a subcategory.
auditpol /list /subcategory:*	List all categories and subcategories by name/GUID.
auditpol /backup /file:audit.csv	Back up audit policy to a CSV file.
auditpol /restore /file:audit.csv	Restore audit policy from a backup.

Subcommand	Purpose
auditpol /clear /y	Reset per-user audit policy.
auditpol /remove /allusers	Remove per-user audit policy for all users.
auditpol /resourcesacl	Configure resource SACLs required for object-access auditing.

4.15 certutil — selected operations

A large certificate-services toolkit; the most commonly used verbs are listed here.

Subcommand	Purpose
certutil -dump file.cer	Decode and display a certificate/CRL in detail.
certutil -verify file.cer	Verify a certificate chain and validity.
certutil -encode/-decode in out	Base64-encode or decode a file (PEM).
certutil -hashfile file SHA256	Compute MD5/SHA1/SHA256 hash of a file.
certutil -urlcache -split -f url	Download a URL through the CryptoAPI cache.
certutil -addstore Root file.cer	Add a certificate to a store (Root, My, CA...).
certutil -delstore -store	Delete or list certificates in a store.
certutil -ping	Test whether the Active Directory Certificate Services service responds.

4.16 nslookup — interactive mode

Start nslookup with no arguments to enter interactive mode; set commands control query behavior.

Subcommand	Purpose
server name / lserver name	Switch the default DNS server (lserver uses the initial server).
root	Point the default server at the DNS root.
set type=A AAAA MX TXT NS SOA CNAME ANY	Choose the record type to query (alias set querytype).
set debug / set d2	Show full response packets / additional raw debugging.
set domain / set srchlist	Set the default domain and search list.
set recurse / set norecurse	Toggle recursive queries.
set retry / set timeout / set port	Tune retries, timeout and server port.
set vc	Force TCP (virtual circuit) instead of UDP.
set class=IN CHAOS	Set query class.
ls domain	List records in a DNS zone (when zone transfer is allowed).
view file	Sort and view a file produced by ls.
finger name / exit	Finger a contact in current context, or leave nslookup.

4.17 ftp — interactive commands

Invoke with `ftp host`, or run `ftp` and use `open`. Script sessions with `ftp -s:file host`. Commands may be abbreviated.

Subcommand	Purpose
<code>open host [port] / close / disconnect</code>	Connect to, or close, an FTP server.
<code>user name</code>	Supply user credentials after connecting.
<code>ascii / binary (bi)</code>	Switch text or image transfer mode (use binary for non-text files).
<code>get (recv) / put (send)</code>	Download or upload one file.
<code>mget / mput</code>	Transfer multiple files matching wildcards.
<code>mdelete / mkdir / rmdir / rename</code>	Manage remote files and directories.
<code>cd / lcd / pwd</code>	Change remote directory, local directory, or print remote directory.
<code>dir / ls / mls / mdir</code>	List remote directory contents in long or short form.
<code>append / delete</code>	Append to a remote file or delete one.
<code>bell / hash / prompt</code>	Sound on completion, show # progress marks, toggle multi-file prompts.
<code>glob</code>	Toggle wildcard expansion for <code>mget/mput/mdelete</code> .
<code>literal (quote) command</code>	Send a raw FTP protocol command.
<code>status / trace / verbose / debug</code>	Show status, packet tracing and verbose replies.
<code>type [A I]</code>	Show or set transfer type.
<code>bye (quit)</code>	End the session and exit.

4.18 telnet — interactive mode

Enable the Telnet Client optional feature first (`optionalfeatures` or `dism`). Press the telnet escape key (default `CTRL+]`) to enter command mode.

Subcommand	Purpose
<code>open host [port]</code>	Connect to a host (optionally a specific port).
<code>close</code>	Close the current connection but stay in telnet.
<code>quit</code>	Close and exit the telnet client.
<code>display</code>	Show current operating parameters.
<code>set / unset option</code>	Set or clear options such as <code>LOCAL_ECHO</code> , <code>TERM</code> and escape character.
<code>send sequence</code>	Send special sequences (<code>ao</code> , <code>ayt</code> , <code>esc</code> , <code>brk</code> , <code>ip</code>).
<code>mode console character</code>	Switch console or character mode.
<code>status</code>	Show connection status.
<code>? / help</code>	List telnet commands.

5. PowerShell Essentials

5.1 Cmdlet Structure and Discovery

PowerShell uses Verb-Noun cmdlets piped as objects: Verb-Noun -Parameter value. Discovery trio: Get-Command finds commands, Get-Help explains them (add -Full or -Online), and Get-Member shows an object's properties and methods. \$_ (alias \$PSItem) is the current pipeline object. Windows PowerShell 5.1 is built in; cross-platform PowerShell 7+ runs as pwsh.

Parameters start with a hyphen and support tab completion; common parameters include -WhatIf (preview) and -Confirm (approve each change). Execution policy controls whether scripts run: see Get-ExecutionPolicy and Set-ExecutionPolicy RemoteSigned -Scope CurrentUser. The ISE and modern editors provide IntelliSense; in a console, Ctrl+Space triggers parameter completion.

5.2 Core Cmdlets by Category

Help, discovery and aliases

Cmdlet	Purpose
Get-Command	Find commands by name, verb, noun or module (-Verb, -Noun, -Module, -CommandType).
Get-Help cmd -Full	Show help with parameters and examples; Update-Help downloads help files.
Get-Member	List the properties and methods of objects coming through the pipeline.
Get-Alias / New-Alias	Resolve or define command aliases (gal, nal).
about_* topics	Conceptual help, e.g. help about_Comparison_Operators.
\$PSVersionTable	Show PowerShell edition and version.

Locations, drives and files

Cmdlet	Purpose
Get-Location / Set-Location	Print or change location (aliases pwd, cd, sl, chdir).
Get-ChildItem	List items (dir, ls, gci); -Recurse, -Filter, -File, -Directory, -Hidden.
Push-Location / Pop-Location	Save and return to locations (pushd/popd).
New-Item -ItemType	Create files, directories, junctions or symbolic links (ni, md).
Copy-Item / Move-Item	Copy or move files and trees (cp/mv; -Recurse, -Force).
Remove-Item	Delete files/folders (rm, del, erase, ri; -Recurse -Force).
Rename-Item	Rename an item (ren, rni).

Cmdlet	Purpose
Get-Content / Set-Content / Add-Content	Read, overwrite or append text (cat, type, gc; -Tail, -Wait).
Out-File / Tee-Object	Write pipeline output to a file or split it to file and screen.
Test-Path / Resolve-Path / Join-Path	Check existence, expand or build paths safely.
Get-Acl / Set-Acl	Read or replace an access control list.
Get-ItemProperty / Set-ItemProperty	Read or write file (and registry) attributes and properties.

Text, objects and data

Cmdlet	Purpose
Select-String	Regex/literal search in files and strings (sls), the grep/findstr equivalent.
Where-Object	Filter pipeline objects (alias ?, where).
Select-Object	Choose properties, first/last N rows or unique objects (select).
Sort-Object / Group-Object	Order or group objects (sort, group).
ForEach-Object	Process each object (alias %, foreach) with process blocks.
Measure-Object	Count, sum, average, min/max, line/word/char counts.
Compare-Object	Diff two object sets (diff), the fc equivalent.
Format-Table / Format-List	Shape console output (ft, fl).
Export-Csv / Import-Csv / ConvertTo-Csv	Write/read CSV data.
ConvertTo-Json / ConvertFrom-Json	Serialize and parse JSON.
Export-Clixml / Import-Clixml	Round-trip typed objects as XML.

Processes and services

Cmdlet	Purpose
Get-Process	List processes with CPU/memory/threads (ps, gps).
Start-Process	Launch executables; -Wait, -Verb RunAs elevates, -NoNewWindow.
Stop-Process	Terminate by name or ID (kill, spps); -Force.
Wait-Process / Debug-Process	Wait for exit or attach a debugger.
Get-Service	List services and status (gsv).
Start/Stop/Restart/Suspend-Service	Control service lifecycle.
Set-Service / New-Service	Change startup type/account or register a service.

Computer, system and events

Cmdlet	Purpose
Get-ComputerInfo	Consolidated OS, BIOS, CPU, memory and domain inventory.
Get-CimInstance / Invoke-CimMethod	Modern WMI access (gcim), e.g. Win32_OperatingSystem, Win32_Process.
Get-HotFix	List installed updates (qfe).

Cmdlet	Purpose
Get-WinEvent	Query event logs with -FilterHashtable (replaces Get-EventLog).
Get-LocalUser / New-LocalUser	Manage local user accounts and passwords.
Get-LocalGroup / Get-LocalGroupMember	Inspect local groups and membership.
Restart-Computer / Stop-Computer	Reboot or shut down, locally or remotely (-Force).
[Environment]::GetEnvironmentVariable	Read user/machine/process environment variables (\$env:NAME for process).

Networking

Cmdlet	Purpose
Test-Connection	ICMP ping as objects; -Count, -Quiet returns a boolean.
Test-NetConnection	TCP port test, route and adapter diagnostics (tnc).
Resolve-DnsName	DNS lookup for any record type.
Get-NetAdapter / Disable/Enable-NetAdapter	List and control network adapters.
Get-NetIPAddress / New-NetIPAddress	Read or assign IP addresses.
Get-NetIPConfiguration	Adapter/IP/DNS/Gateway summary (the ipconfig equivalent).
Set-DnsClientServerAddress	Configure DNS servers on an interface.
Get-NetRoute / New-NetRoute	Read or add routes.
Get-NetFirewallRule / New-NetFirewallRule	Query or create Windows Firewall rules.
Invoke-WebRequest	HTTP client (iwr; alias curl on Windows PowerShell).
Invoke-RestMethod	Call REST/JSON APIs (irm).

Registry, jobs and remoting

Cmdlet	Purpose
HKLM: / HKCU: drives	Browse the registry like a file system with Get-ChildItem and *-ItemProperty.
Start-Job / Get-Job / Receive-Job	Run background jobs and collect their output.
Invoke-Command -ComputerName	Run a script block on remote machines over WinRM (icm).
Enter-PSSession / Exit-PSSession	Interactive remote PowerShell session (etsn).
New-PSSession / Remove-PSSession	Persistent remoting sessions.
Enable-PSRemoting	Configure WinRM remoting on the local machine.

Modules, packages and output

Cmdlet	Purpose
Get-Module / Import-Module	List loaded/available modules or load one.
Find-Module / Install-Module	Discover and install modules from the PowerShell Gallery.
Update-Module / Save-Module / Uninstall-Module	Maintain installed modules.
Get-Package / Find-Package / Install-Package	Work with package providers (msi, nuget, winget provider).

Cmdlet	Purpose
Write-Output / Write-Host	Emit pipeline objects vs. write console-only text.
Write-Verbose / Write-Warning / Write-Error	Structured message streams controlled by -Verbose and try/catch.
\$? and \$LASTEXITCODE	Success flag of the last command and the exit code of the last native .exe.

5.3 CMD-to-PowerShell Equivalent Map

PowerShell defines aliases for familiar commands and provides native cmdlets for classic tasks. Classic console commands (ipconfig, netstat, net, robocopy and so on) also run unchanged inside PowerShell.

Command Prompt	PowerShell equivalent	Note
dir	Get-ChildItem	aliases dir, ls, gci
cd / chdir	Set-Location	aliases cd, sl
md / mkdir	New-Item -ItemType Directory	alias md, ni
type / cat	Get-Content	aliases cat, gc, type
copy / xcopy / robocopy	Copy-Item	robocopy still preferred for large mirror jobs
move	Move-Item	alias mv
del / erase	Remove-Item	aliases del, erase, rm, ri
ren	Rename-Item	alias ren, rni
cls	Clear-Host	alias clear, cls
echo	Write-Output	Write-Host is console-only
findstr / find	Select-String	alias sls
fc	Compare-Object	alias diff
set / setx	\$env:Name / [Environment]::SetEnvironmentVariable	session vs persistent scope
tasklist	Get-Process	alias ps, gps
taskkill	Stop-Process	alias kill; -Force to force
ipconfig	Get-NetIPConfiguration	ipconfig still works inside PowerShell
ping	Test-Connection	-Quiet returns \$true/\$false
tracert	Test-NetConnection - TraceRoute	per-hop route information
nslookup	Resolve-DnsName	structured record objects
netstat	Get-NetTCPConnection	state, local/remote address and owning process
where	Get-Command	locate executables and cmdlets
systeminfo	Get-ComputerInfo	richer object inventory
net user	Get-LocalUser / net.exe still available	domain users: ActiveDirectory module
net use	New-PSDrive -PSProvider FileSystem	persist with -Persist
schtasks	Get-ScheduledTask / Register-ScheduledTask	ScheduledTasks module
reg query	Get-ItemProperty HKLM: HKCU:	registry as a drive

Command Prompt	PowerShell equivalent	Note
shutdown /r	Restart-Computer	-Force closes blocking apps
help / command /?	Get-Help	help is an alias for Get-Help

6. Task-Oriented Quick Index

Look up a goal in the left column and use the command on the right. Replace placeholder values (host names, paths, SSIDs) with your own.

Task	Command
Find current directory and full path	cd ; echo %CD%
List all files recursively, bare names	dir /s /b
Find files modified in the last 7 days and delete them	forfiles /p C:\logs /d -7 /c "cmd /c del @path"
Search text inside files (regex)	findstr /s /i /r /n "pattern" *.txt
Show Wi-Fi passwords saved on the PC	netsh wlan show profiles ; netsh wlan show profile name="SSID" key=clear
Show full IP/DNS/DHCP configuration	ipconfig /all
Release and renew a DHCP lease	ipconfig /release ; ipconfig /renew
Flush the DNS resolver cache	ipconfig /flushdns
Ping with a fixed count	ping -n 4 hostname
Trace the network path to a host	tracert -d hostname (or pathping hostname)
See which process owns a port	netstat -ano findstr :8080 -> tasklist /fi "PID eq <pid>"
Test whether a TCP port is open	Test-NetConnection host -Port 443 (PowerShell)
Look up DNS records	nslookup -type=mx example.com
Map and remove a network drive	net use Z: \\server\share ; net use Z: /delete
List running processes / services per process	tasklist ; tasklist /svc
Force-close a program	taskkill /f /im program.exe /t
List and control services	sc query state= all ; sc start stop config Name
Create a scheduled daily task	schtasks /create /tn Job /tr C:\job.bat /sc daily /st 02:00
Check and repair system files	sfc /scannow
Repair the Windows component store	DISM /Online /Cleanup-Image /RestoreHealth
Check a disk for errors	chkdsk C: /f /r
Defragment/optimize a volume	defrag C: /O
Manage disk partitions	diskpart (list disk, select disk n, clean, create partition primary, format quick)
Convert FAT32 to NTFS without data loss	convert D: /fs:ntfs
Show OS version, uptime and hotfixes	systeminfo ; wmic os get LastBootUpTime
Show who is logged on and privileges	whoami /all
Add a local user / local admin	net user Alice P@ssw0rd /add ; net localgroup administrators Alice /add
Take ownership and reset permissions	takeown /f C:\folder /r /d y ; icacls C:\folder /grant Administrators:F /t

Task	Command
Encrypt / decrypt with EFS	cipher /e folder ; cipher /d folder
Securely wipe free space	cipher /w:C:
Compress (NTFS) a folder	compact /c /s:C:\folder
Robust mirror copy with logging	robocopy C:\src D:\dst /MIR /Z /R:2 /W:2 /LOG:copy.log
Create a symbolic link / junction	mklink /D link target ; mklink /J junction target
Substitute a drive letter for a folder	subst Z: C:\some\folder ; subst Z: /d
Shut down / restart / abort	shutdown /s /t 0 ; shutdown /r /t 0 ; shutdown /a
Restart a remote machine	shutdown /r /m \\host /t 30 /c "message"
Back up with Windows Server Backup	wbadmin start backup -backuptarget:E: -include:C: -quiet
Check VSS writers during backup issues	vssadmin list writers
Check BitLocker status	manage-bde -status
Show installed drivers	driverquery /v
Read the latest 20 system event log entries	wevtutil qe System /c:20 /rd:true /f:text
Capture network packets in-box	pktmon start --capture --comp all -f trace.etl ; pktmon stop
Compute a file hash	certutil -hashfile file.iso SHA256
Base64-encode a file	certutil -encode in.bin out.b64
Download a file from the command line	curl.exe -L -o file.zip https://example.com/file.zip
Extract a zip/tar archive	tar -xf archive.zip -C C:\out
Generate an SSH key and connect	ssh-keygen -t ed25519 ; ssh user@host
Install or enter a Linux distribution	wsl --install -d Ubuntu ; wsl
Install or upgrade software packages	winget search name ; winget install --id Publisher.Package ; winget upgrade --all
Set a persistent environment variable	setx JAVA_HOME "C:\Java" /m (machine-wide, new windows)
Pause a script for 15 seconds	timeout /t 15 /nobreak
Send output to the clipboard	ipconfig /all clip
Change console code page to UTF-8	chcp 65001
Show Windows licensing status	slmgr /dlv (script) ; slmgr /xpr expiry date

Appendix A. Redirection, Operators and Batch Symbols

Symbol / pattern	Meaning
command > file.txt	Redirect standard output to a file (overwrite).
command >> file.txt	Append standard output to a file.
command 2> err.txt	Redirect error output (stream 2) to a file.
command > out.txt 2>&1	Send success and error output to the same file.
command 2>nul	Discard error output (nul is the null device).
command1 command2	Pipe standard output of command1 into command2.
command1 & command2	Run command2 after command1 regardless of success.
command1 && command2	Run command2 only if command1 succeeds (ERRORLEVEL 0).

Symbol / pattern	Meaning
command1 command2	Run command2 only if command1 fails.
(commands)	Group multiple commands; combine with redirection.
@command	Suppress echoing of that command itself (@echo off for a whole script).
^	Escape the next special character (^ , ^>, ^&, ^^) outside quotes.
* and ?	Wildcards: * any characters, ? one character (in file names).
%VAR%	Expand an environment variable when the line is parsed.
!VAR!	Expand a variable at execution time under setlocal enabledelayedexpansion.
%0 ... %9	Batch script name and positional parameters; %* is all parameters.
%~dp0	Drive and folder path of the running batch script.
%ERRORLEVEL%	Numeric exit code returned by the previous command.
%DATE% %TIME% %RANDOM% %USERNAME% %COMPUTERNAME%	Built-in dynamic environment variables.
"path with space"	Quote paths and arguments containing spaces.
command /?	Display built-in help and switch list for a command.
nul / con	Reserved device names: null device and console.

Appendix B. Run (Win+R) Dialog and System Utility Commands

Press Win+R and type any entry below. .msc files open Microsoft Management Console snap-ins, .cpl files open Control Panel items, and the remaining entries launch system tools directly.

Type this	Opens
cmd / powershell / pwsh	Command Prompt, Windows PowerShell, PowerShell 7
wt	Windows Terminal (when installed)
regedit / regedt32	Registry Editor
services.msc	Services console
devmgmt.msc	Device Manager
diskmgmt.msc	Disk Management
compmgmt.msc	Computer Management (event viewer, users, disks combined)
eventvwr.msc	Event Viewer
taskschd.msc	Task Scheduler
gpedit.msc	Local Group Policy Editor (Pro/Enterprise editions)
secpol.msc	Local Security Policy
lusrmgr.msc	Local Users and Groups
certmgr.msc	Current-user certificate manager

Type this	Opens
wf.msc	Windows Defender Firewall with Advanced Security
printmanagement.msc	Print Management console
fsmgmt.msc	Shared Folders (shares, sessions, open files)
tpm.msc	Trusted Platform Module management
comexp.msc / dcomcnfg	Component Services and DCOM configuration
odbcad32	ODBC Data Source Administrator (32/64-bit variants)
ncpa.cpl	Network Connections
appwiz.cpl	Programs and Features (uninstall)
sysdm.cpl	System Properties (computer name, performance, environment variables)
powercfg.cpl	Power Options
firewall.cpl	Windows Defender Firewall
inetcpl.cpl	Internet Properties
mmsys.cpl	Sound properties
desk.cpl	Display settings / screen resolution
main.cpl	Mouse properties
intl.cpl	Region and language settings
timedate.cpl	Date and Time
colorcpl	Color Management
control	Control Panel (control /name Microsoft.X opens named pages)
msconfig	System Configuration (boot, services, startup, tools)
msinfo32	System Information
dxdiag	DirectX Diagnostic Tool
resmon	Resource Monitor
perfmon	Performance Monitor
verifier	Driver Verifier Manager
sigverif	File Signature Verification
cleanmgr / dfrgui	Disk Cleanup / Optimize Drives
winver	About Windows (edition and build)
slmgr.vbs	Windows Software Licensing (activation) script; /dlv, /xpr, /ato
optionalfeatures	Turn Windows features on or off
taskmgr	Task Manager
explorer	File Explorer (shell:startup opens the Startup folder)
notepad / calc / mspaint / write	Notepad, Calculator, Paint, WordPad
osk / narrator / magnify	On-screen keyboard, Narrator, Magnifier (accessibility)
mstsc	Remote Desktop Connection
mmc	Blank Microsoft Management Console for custom .msc files
ms-settings:	Open the modern Settings app (append page URI, e.g. ms-settings:windowsupdate)

Appendix C. Compatibility Notes and Sources

C.1 Compatibility Notes

- Deprecated commands (at, cacs, bootcfg, ntbackup, rsh, rexec, wmic and similar) still exist on many systems for compatibility but should not be used in new automation; their descriptions name the replacement.
- 16-bit commands (edit, diskcomp/diskcopy for floppies, graftabl and similar) are unavailable on 64-bit Windows.
- Active Directory and server tools (dcdiag, repadmin, netdom, dnscmd, wdsutil, nfs*) require the corresponding role or RSAT tools.
- curl.exe, tar.exe, ssh.exe/scp.exe/sftp.exe/ssh-keygen.exe, wsl.exe and winget.exe are present only on modern Windows 10/11 releases (or after installing the named optional feature).
- Switches and output differ slightly between Windows versions; always confirm with command /? on the target system.

C.2 Sources

Command inventory and semantics were compiled from the official Microsoft Learn documentation:

- Windows commands A–Z reference: <https://learn.microsoft.com/windows-server/administration/windows-commands/windows-commands>
- PowerShell documentation: <https://learn.microsoft.com/powershell/>
- Windows client and networking command references (netsh, pktmon, DISM, OpenSSH, WSL, winget): <https://learn.microsoft.com/windows/>

This reference is descriptive documentation of built-in operating-system capabilities; verify version-specific behavior against the documentation for your exact Windows build.